

PROJECT REPORT :

“IoT-Based Smart Door Lock System Using Fingerprint Authentication and Wi-Fi Alerts”

- **Hardware:** ESP32 DevKit V1, R307 fingerprint sensor, magnetic reed sensor, SG90 servo, buzzer, LEDs, MB102 power supply
- **Software:** Arduino IDE, Embedded C/C++, Wi-Fi, Telegram Bot API

Main Chapters

Chapter 1 — Introduction
Chapter 2 — Problem Statement and Objectives
Chapter 3 — System Requirements and Components
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ABSTRACT

Traditional door-locking systems such as mechanical keys and conventional locks provide basic physical security but offer limited capabilities for user authentication, remote monitoring, and unauthorized-access notification. To address these limitations, an IoT-Based Smart Door Lock System Using Fingerprint Authentication and Wi-Fi Alerts has been developed.

The proposed system uses an ESP32 DevKit V1 as the main microcontroller and integrates an R307 fingerprint sensor for biometric authentication, an SG90 servo motor for electronic locking and unlocking, and a magnetic reed sensor for monitoring the physical state of the door. LEDs and a buzzer provide local visual and audible indications of system status. The ESP32 uses its built-in Wi-Fi capability to communicate with a Telegram Bot API, enabling remote notifications for unauthorized attempts, system lock events, and detected forced door opening.

During operation, a registered fingerprint is verified against the fingerprint templates stored in the sensor's internal memory. If authentication is successful, the servo temporarily unlocks the door and automatically returns to the locked position. For an unauthorized fingerprint, the system activates the warning indicators and sends an alert through Telegram. After three consecutive failed authentication attempts, a temporary lock mode is activated.

CHAPTER 1 — INTRODUCTION

1.1 Background

Security and controlled access are important requirements in homes, offices, laboratories, institutions, and other restricted areas. Conventional mechanical locks generally depend on physical keys, which can be lost, duplicated, or shared. Although electronic access systems provide additional functionality, many basic systems do not combine biometric authentication with physical door monitoring and remote notification.

The development of low-cost microcontrollers and IoT technologies has made it possible to design compact security systems that combine authentication, automation, sensing, and wireless communication.

The proposed project develops an IoT-based smart door lock system in which fingerprint authentication is used to verify the user before granting access. An ESP32 microcontroller processes the authentication result and controls a servo motor responsible for the locking mechanism. A magnetic reed sensor monitors the door state, while LEDs and a buzzer provide immediate local feedback.

The system also uses Wi-Fi to send security-related notifications through Telegram. This allows the user to receive information about unauthorized access attempts and other security events without being physically present near the door.

1.2 Motivation

The project is motivated by the need for an access-control system that can provide:

- Keyless user authentication.
- Controlled automatic door locking and unlocking.
- Detection of unauthorized authentication attempts.
- Detection of unexpected door-state changes.
- Local visual and audible alerts.
- Remote security notifications.
- A low-cost and compact embedded implementation.

By combining these features, the project demonstrates how embedded systems and IoT communication can be integrated into a practical security application.

CHAPTER 2 — PROBLEM STATEMENT AND OBJECTIVES

2.1 Problem Statement

Conventional door-locking mechanisms primarily provide physical access control but do not inherently provide biometric authentication, automatic relocking, intrusion detection, or remote security notifications.

Therefore, there is a need for a compact electronic security system that can authenticate authorized users, control the locking mechanism, identify repeated unauthorized attempts, monitor the physical door state, and provide remote alerts.

2.2 Objectives

The main objectives of the project are:

1. To design an electronic door-locking system using the ESP32 DevKit V1.
2. To implement fingerprint-based authentication using an R307 fingerprint sensor.
3. To control door locking and unlocking using an SG90 servo motor.
4. To detect changes in the physical door state using a magnetic reed sensor.
5. To provide local access-status indication using green and red LEDs.
6. To generate audible alerts using a buzzer.
7. To implement a temporary lock mechanism after repeated failed authentication attempts.
8. To send security notifications through Wi-Fi using the Telegram Bot API.
9. To integrate the complete system into a physical PCB-based prototype.

CHAPTER 3 — SYSTEM REQUIREMENTS AND COMPONENTS

3.1 Hardware Requirements

Component	Purpose
ESP32 DevKit V1	Main controller
R307 Fingerprint Sensor	Biometric authentication
SG90 Servo Motor	Lock/unlock mechanism
Magnetic Reed Sensor	Door-state/forced-entry detection
Green LED	Access granted indication
Red LED	Access denied/system lock indication
Buzzer	Audible warning
MB102 Power Supply Module	External power distribution
Resistors	LED current limiting
PCB	Permanent hardware implementation
Connecting wires	Electrical interconnections

3.2 Software Requirements

1. Arduino IDE
2. Embedded C/C++
3. ESP32 board package
4. Adafruit Fingerprint Sensor Library
5. ESP32Servo Library
6. Wi-Fi library
7. HTTPClient library
8. Telegram Bot API

CHAPTER 4 — SYSTEM DESIGN AND ARCHITECTURE

4.1 System Architecture

The system is divided into four major sections:

Input Section

- R307 Fingerprint Sensor
- Magnetic Reed Door Sensor
- Processing Section
- ESP32 DevKit V1

Output Section

- SG90 Servo Motor
- Green LED
- Red LED
- Buzzer

Communication Section

- ESP32 Wi-Fi
- Telegram Bot API
- Power Section
- MB102 Power Supply Module

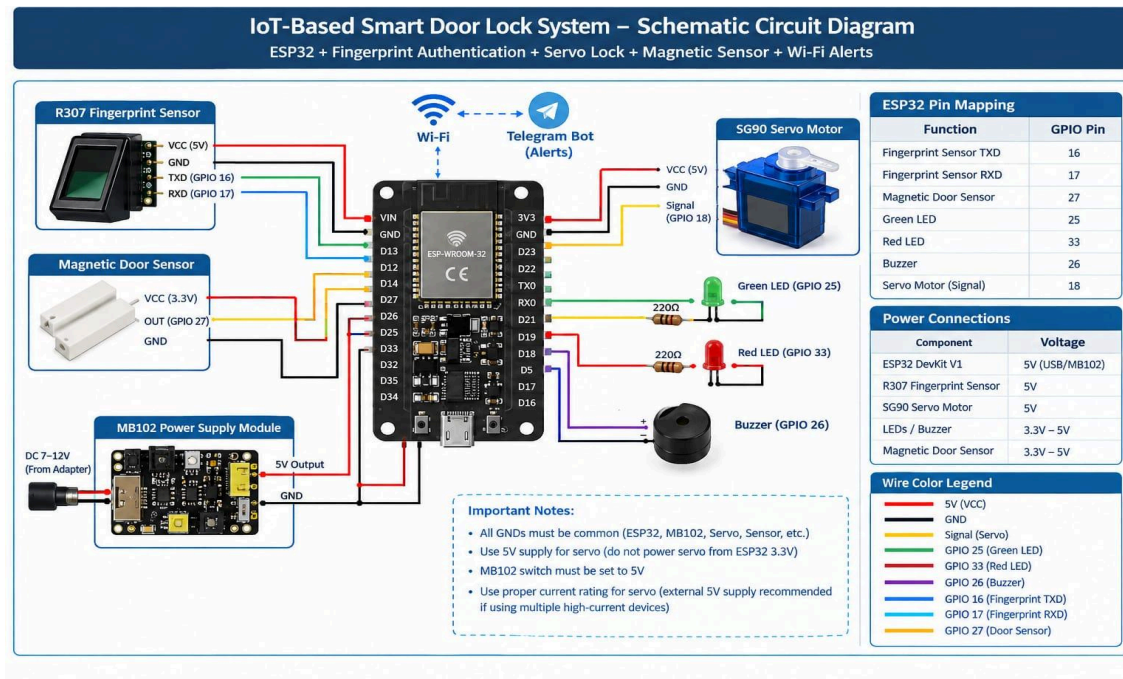


Figure 4.1: System Circuit Diagram

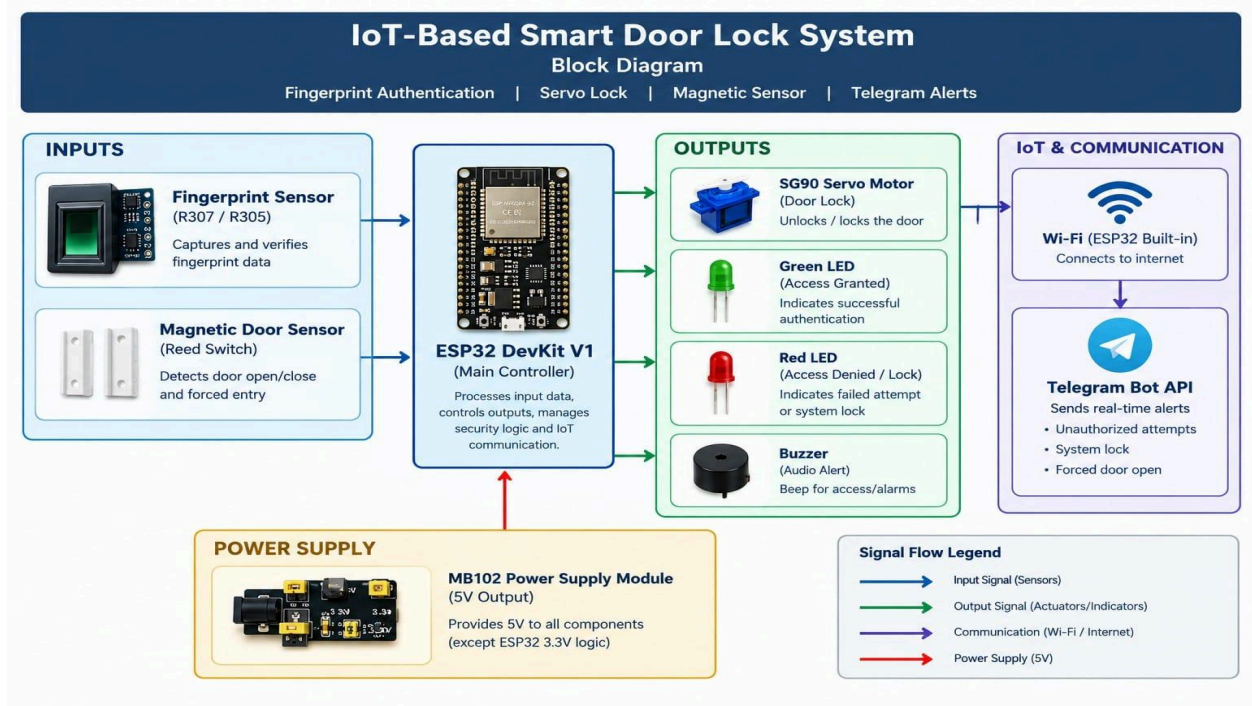


Figure 4.2: Schematic Block Diagram

4.2 Pin Configuration

Component	ESP32 Pin	Function
Fingerprint TXD	GPIO 16	UART2 RX
Fingerprint RXD	GPIO 17	UART2 TX
Magnetic Reed Sensor	GPIO 27	Digital input
Green LED	GPIO 25	Access granted
Red LED	GPIO 33	Access denied/lock
Buzzer	GPIO 26	Audible alert
Servo Signal	GPIO 18	PWM control

NOTE :The ESP32 GPIO assignments were selected based on the available digital, UART, and PWM-capable interfaces and the requirements of the implemented hardware configuration.

CHAPTER 5 — HARDWARE IMPLEMENTATION

5.1 ESP32 DevKit V1

The ESP32 DevKit V1 acts as the central controller of the system. It receives fingerprint authentication data through UART, monitors the magnetic door sensor, controls the servo and indication devices, and manages Wi-Fi communication with the Telegram Bot API.

5.2 R307 Fingerprint Sensor

The R307 fingerprint sensor is used for biometric authentication. During enrollment, the sensor captures the same finger twice, converts the captured images into templates, compares them, and stores the resulting fingerprint model in its internal memory with a unique identification number.

During authentication, the live fingerprint is converted into a template and compared with the stored templates. A matching fingerprint produces the corresponding stored ID, which is then processed by the ESP32.

5.3 SG90 Servo Motor

The SG90 servo motor acts as the actuator of the system. The ESP32 controls its angular position using a PWM signal.

In the implemented system:

0° → Locked position

90° → Unlocked position

After successful authentication, the servo moves to the unlocked position for approximately three seconds and then returns to the locked position.

5.4 Magnetic Reed Sensor

The magnetic reed sensor is used as the second sensor in the system. It detects a change in the physical state of the door.

The sensor is configured as a digital input using the ESP32's internal pull-up configuration. During temporary lock mode, the initial door state is recorded. A subsequent change in this state is interpreted as a possible forced door opening.

5.5 LED and Buzzer Indication

The system uses two LEDs:

- Green LED: Access granted.
- Red LED: Access denied or temporary lock mode.

The buzzer provides an audible warning during unauthorized access attempts and forced-entry detection.

5.6 Power Supply

An MB102 power supply module is used to provide the required external supply for the hardware components. The servo is supplied from the appropriate 5 V supply rather than relying on the ESP32's 3.3 V output.

A common ground is maintained between the ESP32 and connected modules to ensure proper signal reference.

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CHAPTER 6 — SOFTWARE IMPLEMENTATION

6.1 Development Environment

The firmware was developed using the Arduino IDE with ESP32 board support.

The program integrates:

- Fingerprint authentication
- Servo control
- Door-state monitoring
- LED and buzzer control
- Wi-Fi connectivity
- Telegram notifications
- Failed-attempt counting
- Temporary system lock

6.2 Fingerprint Enrollment

The enrollment process is performed before normal system operation.

Steps:

1. A unique fingerprint ID is entered.
2. The user places a finger on the sensor.
3. The sensor captures the first fingerprint image.
4. `image2Tz(1)` converts the first image into a fingerprint template.
5. The user removes the finger.
6. The same finger is placed again.
7. The second image is captured.
8. `image2Tz(2)` converts the second image into another template.
9. `createModel()` compares both templates.
10. If they match, `storeModel(id)` stores the fingerprint model in the sensor's internal memory.

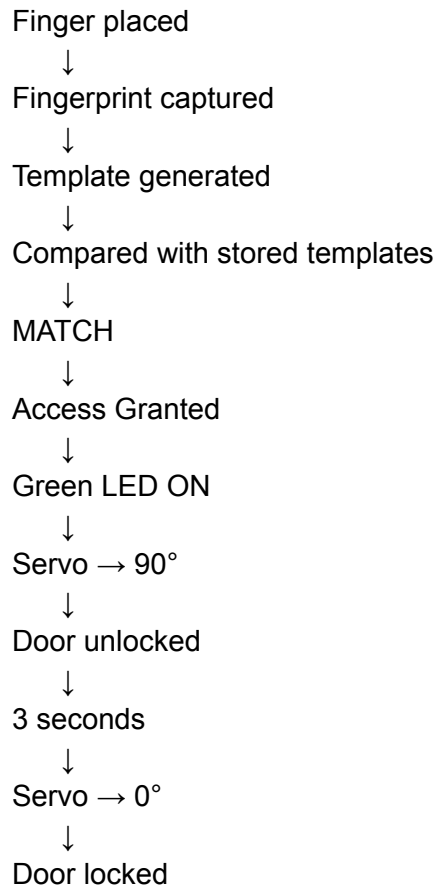
This allows multiple authorized fingerprints to be registered using different IDs.

CHAPTER 7 — WORKING PRINCIPLE

7.1 Normal Operation

When the system starts, the ESP32 initializes the connected devices and establishes Wi-Fi connectivity. The fingerprint sensor is then ready to authenticate users.

Case 1 — Authorized Fingerprint



The failed-attempt counter is also reset after successful authentication.

7.2 Unauthorized Fingerprint

Finger placed



Fingerprint captured



No matching stored template



Access Denied



Red LED + Buzzer



Telegram Alert



Attempt Counter +1

The system records the number of consecutive failed attempts.

7.3 Temporary Lock Mode

After three consecutive failed authentication attempts, the system enters temporary lock mode.

During this period:

- The servo remains in the locked position.
- The red LED remains active.
- The system displays a countdown through the serial monitor.
- A Telegram notification is sent.
- Further authentication processing is temporarily suspended.
- The magnetic sensor monitors for a change in door state.

The temporary lock period in the implemented prototype is 10 seconds.

After the timeout, the system returns to normal operation and the failed-attempt counter is reset.

7.4 Forced Door Opening Detection

When temporary lock mode begins, the ESP32 stores the current state of the magnetic sensor.

If the sensor state changes during the lock period:

Door State Changed



Possible Forced Entry



Buzzer Alarm



Telegram Alert

CHAPTER 8 — WI-FI AND TELEGRAM ALERT SYSTEM

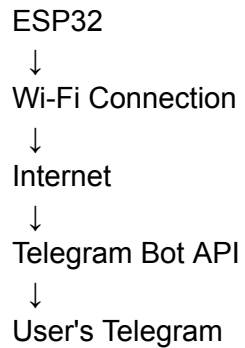
The ESP32 uses its built-in Wi-Fi functionality to connect the system to the internet.

The Telegram Bot API is used to transmit security notifications.

The system can generate notifications for:

1. Unauthorized fingerprint attempt.
2. Temporary system lock after repeated failures.
3. Forced door opening during lock mode.

The basic communication process is:



This allows security events to be communicated remotely rather than relying only on local LEDs and buzzer indications.

CHAPTER 9 — TESTING AND RESULTS

Table 8.1 — Testing and Results
IoT-Based Smart Door Lock System Using Fingerprint Authentication and Wi-Fi Alerts

Sr. No	Test Case	Input / Condition	Expected Result	Observed Result	Status
1	System Initialization	Power ON	ESP32 initializes all modules	System initialized successfully	Pass
2	Fingerprint Authentication	Registered fingerprint	Fingerprint recognized; access granted	Fingerprint recognized successfully	Pass
3	Door Unlocking	Authorized fingerprint	Servo rotates to 90°	Servo unlocked the door	Pass
4	Automatic Relocking	After 3 seconds	Servo returns to 0°	Door locked automatically	Pass
5	Unauthorized Access	Unregistered fingerprint	Access denied; warning activates	Red LED and buzzer activated	Pass
6	Wi-Fi Notification	Unauthorized attempt	Telegram alert sent	Telegram alert received	Pass
7	Failed Attempt Counter	Repeated failed attempts	Counter increments	Counter increased correctly	Pass
8	Temporary Lock	Three failed attempts	10-second lock mode activates	System entered lock mode	Pass
9	Lock Mode Indication	Lock mode active	Red LED and buzzer indicate lock	Red LED and buzzer activated	Pass
10	Forced Door Opening	Door changes during lock	Forced entry detected	Forced entry detected	Pass
11	Forced Entry Alert	Forced opening detected	Buzzer + Telegram alert	Alert generated successfully	Pass
12	Lock Timeout	10 seconds completed	System returns to normal	System returned to normal	Pass
13	Attempt Counter Reset	Valid fingerprint after failures	Counter resets to zero	Counter reset successfully	Pass

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CHAPTER 10 — ADVANTAGES, LIMITATIONS AND APPLICATIONS

10.1 Advantages

- Keyless biometric authentication.
- Automatic locking after access.
- Separate physical door-state monitoring.
- Local LED and buzzer indication.
- Remote security notifications.
- Temporary lock after repeated failed attempts.
- Compact ESP32-based implementation.
- Can be extended with additional IoT functionality.

10.2 Limitations

- Wi-Fi-based alerts require an available internet connection.
- The prototype depends on an external power supply for reliable operation of higher-current components such as the servo.
- The current implementation uses Telegram as the remote notification platform.
- The prototype is designed primarily as an academic demonstration and would require additional mechanical and electrical protection for commercial deployment.

10.3 Applications

The system can be adapted for:

- Residential doors
- College laboratories
- Computer laboratories
- Offices
- Restricted equipment rooms
- Small storage areas
- Workshops
- Institutional access-control systems

CHAPTER 11 — FUTURE SCOPE

The project can be further developed by incorporating:

1. Mobile Application :

A dedicated mobile application could provide real-time access notifications, system status, and remote monitoring.

2. Cloud-Based Access Logs :

Authentication events could be stored in a cloud database with timestamps and user IDs for historical analysis.

3. Multi-Factor Authentication :

Fingerprint authentication could be combined with RFID, PIN, OTP, or another authentication method.

4. Battery Backup :

A rechargeable backup power system could maintain operation during temporary power failures.

5. Tamper Detection :

Additional sensors could detect attempts to physically tamper with the enclosure or locking mechanism.

6. Improved Access Management :

An administrator interface could allow authorized fingerprints to be added, deleted, or managed without directly modifying the firmware.

7. Advanced Monitoring :

A web or mobile dashboard could display access logs, failed attempts, door status, and system alerts.

8. Improved Security :

Secure credential storage, encrypted communication, and improved device-level security could be incorporated for a deployment-oriented version.

CHAPTER 12 — CONCLUSION

The IoT-Based Smart Door Lock System Using Fingerprint Authentication and Wi-Fi Alerts successfully demonstrates the integration of biometric authentication, embedded control, physical door monitoring, actuator control, and IoT communication.

The ESP32 serves as the central controller, processing input from the fingerprint and magnetic door sensors while controlling the servo motor, LEDs, and buzzer. Fingerprint authentication provides keyless access control, while the servo enables automatic locking and unlocking. The magnetic reed sensor adds physical door-state monitoring and enables forced-entry detection during temporary lock mode.

The implementation also demonstrates remote security notification through Wi-Fi and the Telegram Bot API. The three-attempt temporary lock mechanism provides an additional response to repeated unauthorized authentication attempts.

Overall, the developed prototype provides a practical demonstration of how embedded systems and IoT technologies can be combined to create a smart access-control system. The modular architecture also provides a foundation for future enhancements such as cloud logging, mobile applications, multi-factor authentication, and improved security mechanisms.

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